

Unit 6: Practical Skills in Biology II

IA2 compulsory unit

Externally assessed

Unit description

Introduction

Students are expected to develop a wide knowledge and understanding of experimental procedures and techniques throughout the whole of their International Advanced Level course. They are expected to carry out the core practicals and other recommended practical investigations and experiments while they study units 4 and 5.

Students are expected to become aware of how these techniques and procedures might be used to investigate interesting biological questions.

This unit will assess students' ability to apply their knowledge and understanding of experimental procedures and techniques, and their ability to plan whole investigations, analyse data and to evaluate their results and experimental methodology.

Practical skills identified for assessment

- Solve problems set in practical contexts.
- Apply scientific knowledge to practical contexts.
- Comment on experimental design and evaluate scientific methods.
- Present data in appropriate ways.
- Evaluate results and draw conclusions with reference to measurement uncertainties and errors.
- Identify variables, including those that must be controlled.
- Plot and interpret graphs.
- Process and analyse data using appropriate mathematical skills (see *Appendix 6: Mathematical skills and exemplifications*).
- Know and understand how to use a wide range of apparatus, materials and techniques safely, appropriate to the knowledge and understanding in this specification.
- Plan an investigation to test a hypothesis.

Practical skills to be developed through teaching and learning

- Apply investigative approaches and methods to practical work.
 - Use a range of practical equipment and materials safely and correctly.
 - Follow written instructions.
 - Make and record observations.
 - Present information and data in a scientific way.
 - Use appropriate software and tools to collect and process data.
 - Use online and offline research skills, including websites, textbooks and other printed scientific sources of information.
 - Cite sources of information correctly.
 - Use a wide range of experimental and practical instruments, equipment and techniques appropriate to the knowledge and understanding in this specification.
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